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MEMORY MANAGEMENT WITH IMPLICIT IDENTIFICATION OF CRYPTOGRAPHIC KEYS USING ERROR CORRECTION DATA

Abstract

Disclosed systems and techniques involve storage of encrypted data in memory pages that may include units stored with different cryptographic keys. Data may be stored with error correction data that implicitly encodes an identification of a key (key selector) without additional memory being allocated to explicit storage of the key selector. During data retrieval, the key selector is recovered from error correction data by processing multiple instances of extended data in which the data is combined with various possible key selectors.

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Background/Summary

RELATED APPLICATIONS [0001] This is a national stage application under 35 U.S.C. 371 of International Application PCT/US23/24375, filed Jun. 2, 2023, which claims benefit to U.S. Provisional Patent Application No. 63/470,295, filed Jun. 1, 2023, which claims benefit to U.S. Provisional Patent Application No. 63/348,841, filed Jun. 3, 2022 the contents of both are incorporated by reference in their entirety herein.

TECHNICAL FIELD

[0002] The instant disclosure pertains to storage and retrieval of cryptographically encrypted data in computing systems and applications deploying multiple cryptographic keys and, more specifically, to management of memory pages shared among multiple hosts, to allocation of private copies of shared pages to individual hosts, and to using allocated pages for storing and retrieval of data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various implementations of the disclosure.

[0004] FIG. 1 is a block diagram illustrating an example system architecture in which implementations of the present disclosure may operate.

[0005] FIG. 2A is a schematic diagram illustrating example operations of error correction encoding with implicit storage of key identification using error correction data, in accordance with some implementations of the present disclosure.

[0006] FIG. 2B illustrates schematically a memory page that stores data encrypted with different keys, in accordance with some implementations of the present disclosure.

[0007] FIG. 2C is a schematic diagram illustrating example operations of error correction decoding with recovery of an implicitly stored key selectors, in accordance with some implementations of the present disclosure.

[0008] FIG. 3 illustrates an example architecture of a cryptographic accelerator capable of supporting write and read memory operations using implicitly stored identification of cryptographic keys, in accordance with some implementations of the present disclosure.

[0009] FIG. 4 is a flow diagram of an example method of performing a write operation with implicit storage of cryptographic key identification using error correction data, in accordance with some implementations of the present disclosure.

[0010] FIG. 5 is a flow diagram of an example method of performing a read operation with recovery of implicitly stored, using error correction data, cryptographic key identification, in accordance with some implementations of the present disclosure.

[0011] FIG. 6 depicts a block diagram of an example computer system operating in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0012] Aspects of the present disclosure are directed to efficient management of memory resources protected by multiple cryptographic keys. In various computing architectures, such as virtual machine (VM) environments, memory pages allocated to multiple VMs may initially store the same data. Consequently, instead of allocating separate physical memory pages to different VMs, a

single shared physical memory page may be initialized (e.g., populated with zeros or any other initialization data) and made available to the VMs, each VM addressing the shared physical memory page using a VM-specific logical address. A Memory Management Unit (MMU) may map these logical addresses to the same shared physical memory page. Requests to read data from the shared memory page return the same data to the VMs. A request to write data by an individual VM causes the MMU to intercept the write request, copy the contents of the shared memory page to a new page, and perform the write operation by modifying the contents of the new page (“Copy-on-Write”). The new private page is then permanently associated with the individual VM (e.g., for the duration of the VM's session). This alleviates the need for the computing system (e.g., a hypervisor) to allocate many duplicate physical memory pages but allows the MMU to flexibly respond to increases in memory usage in the system.

[0013] In an Allocation-on-Write variation of the Copy-on-Write technique, a pool of initialized pages (“zero pages”) is created (e.g., at the start of a VM session) and the subsequent write requests cause the MMU to select an already initialized page from the pool and write the data to the selected page. This reduced the time needed for initialization of a new page from several microseconds (for the standard Copy-on-Write technique) to several hundreds of nanoseconds.

[0014] In modern trusted VM architectures, memory space of individual VMs is often encrypted with a unique cryptographic key (e.g., a temporary ephemeral AES (Advanced Encryption Standard) key) associated with a specific VM and not accessible to other VMs. Without a correct key, data stored in a memory space of one VM would appear scrambled to another VM, e.g., a page of all zeros would appear as a page filled with non-zero data. This precludes pre-initializing a pool of zero pages available to multiple VMs since upon allocation of a page to a certain VM the new data stored in the allocated page has to be encrypted with that VM's unique key. Maintaining separate pools for different VMs (e.g., thousands of VMs) is inefficient as it may be difficult or impossible to predict accurately, at the time of initialization, how much memory each VM will use. A possible solution that allows the MMU to maintain a single pool of memory pages may include storing a key selector with each unit of data (e.g., a cache line, block, etc.) to indicate whether a key to be used in decrypting that unit is an initialization key or a VM key. For example, units of the memory page still storing initialization data may be encrypted with the initialization key while units already modified by the VM may be encrypted with a key of that VM. Even though such key selectors may be 1-bit selectors (e.g., bit value 0 indicating that the unit of data is encrypted with the initialization key and bit value 1 indicating that the unit of data is encrypted with the VM key), storing such key selectors directly in memory, however, consumes valuable space and is detrimental to system's performance and cost.

[0015] Aspects and implementations of the present disclosure address these and other shortcomings of the existing technology by describing systems and techniques for enabling selective encryption of data using multiple keys and storage of the encrypted data in a single memory page, without explicitly storing key selectors with the data. In one example implementation, tracking of key selectors (e.g., taking values 0 or 1, in the instance of two keys) may be performed by implicitly storing key selectors using error correction (EC) data. More specifically, an EC code used by a memory controller may generate and store k EC bits (also referred to as parity bits herein) for each n bits of data and store the resulting $n+k$ bit codeword in system memory. After a read operation, the EC bits (also referred to as EC data herein) may be used to identify locations of bit errors that have occurred during write operations, storage of the data, and/or read operations. For example, a Reed-Solomon (RS) code may be capable of correcting up to $k/2$ errors located anywhere among the $n+k$ bits of the codeword (or up to k errors located in known positions). Implicit storing of the key selector during a write operation may include expanding (e.g., concatenating) n bits of data encrypted with an appropriate key (e.g., initialization key or VM key) with one key selector bit identifying the correct key (or N key selector bits, if selection is to be performed from $2^{\text{sup}}N$ keys), $\text{data.sub.n.fwdarw.data.sub.n|KeyID.sub.N}$, and a subsequent application of the EC code to

the expanded data to compute k EC bits: $EC(data.sub.n|KeyID.sub.N).fwdarw.ECbits.sub.k$. A codeword is then formed by combining the n -bit data with k EC bits while omitting the key selector: $CW=data.sub.n|ECbits.sub.k$. The codeword CW is stored in system memory. During a read operation, the retrieved codeword is transformed into two expanded codewords (or $2 \cdot sup.N$ expanded codewords, if selection is to be performed from $2 \cdot sup.N$ keys), $CW.fwdarw$.

custom-character.sub.0, custom-character.sub.1, where

[00001]

$$0 = data_n \cdot \text{Math. KeyID}_{init} \cdot \text{Math. ECbits}_k, \quad 1 = data_n \cdot \text{Math. KeyID}_{VM} \cdot \text{Math. ECbits}_k,$$

and two (or $2 \cdot sup.N$) instances of an EC decoder may be applied (e.g., in parallel or sequentially) to custom-character.sub.0 and custom-character.sub.1 separately. The correct key selector may be determined based on the output of the EC decoding operations. For example, if $data.sub.n$ and $ECbits.sub.k$ have m combined errors such that $m < t$, where t is the maximum number of correctable errors (e.g., $t = k/2$ for RS codes), application of the EC decoder to the expanded codeword having a correct key selector will detect m errors and application of EC decoder to the expanded codeword having a wrong key selector will detect $m+1$ errors with the extra error pointing to the incorrect key selector. Both instances of the EC decoder will, therefore, identify both the data and the correct key selector. If $data.sub.n$ and $ECbits.sub.k$ have $m=t$ combined errors, application of the EC decoder to the expanded codeword having the correct key selector will recover the data and application of the EC decoder to the expanded codeword having the wrong key selector will detect an unrecoverable failure. The instance of the successful EC decoder will, therefore, identify the correct key selector. If $data.sub.n$ and $ECbits.sub.k$ have $m > t$ combined errors, application of EC decoders to both codewords may return an unrecoverable failure (although may sometimes, e.g., probabilistically, still recover the correct data and key selector).

[0016] Various implementations and modifications of these and other EC encryption techniques that enable storage of data encrypted with multiple keys in a single memory page are described below. The advantages of the disclosed implementations include but are not limited to efficient management of memory resources in trusted computing environments.

[0017] FIG. 1 is a block diagram illustrating an example system **100** in which implementations of the present disclosure may operate. The example system **100** may support VMs **102-1**, **102-2**, **102-3**, **102-4** etc. The number of VMs **102-j** need not be limited. VMs **102-j** may be supported by a one or more hypervisors (VM monitors) **104**. VMs **102-j** may be provided to users (including local users and/or remote users) who may operate the respective VMs as if VMs were local to the users. In some implementations, VMs **102-j** may be accessed via any computing device, such as a desktop computer, a tablet, a smartphone, a server (local or remote), a thin/lean client, and the like. VMs **102-j** may support various software stacks (SW) **106-1**, **106-2**, **106-3**, **106-4**, etc., which may include guest operating systems, and one or more applications. Applications supported by VMs **102-j** may include machine-learning applications, graphics applications, computational applications, cryptographic applications (such as authentication, encryption, decryption, secure storage applications, etc.), embedded applications, external applications, or any other types of applications. Applications and other software programs operating on VMs **102-1** may be executed on a multi-core processing device, e.g., CPU **140**. CPU **140** may have multiple cores **142-1**, **142-2**, etc. Each core **142-k** may execute instructions of one or more VMs **102-j** and may include a corresponding memory management unit **144-1**, **144-2**, etc.

[0018] VMs **102-j** and SW **106-j** running thereon may be protected with cryptographic keys. VMs **106-j** may store (write) data in memory **130** and retrieve (read) data from memory **130**. Store and retrieve memory operations may be facilitated by memory controller **108**. In some implementations, memory operations may be further facilitated by a cryptographic accelerator **110**. Cryptographic accelerator **110** may include an encryption engine **112** that generates, assigns, and manages encryptions keys of VMs **102-j** (and hypervisor **104**). More specifically, encryption

engine **112** may maintain VM keys **114** that protect data generated by individual VMs **102-j** and an initialization key **116** that protects data shared among multiple VMs **102-j**. Although multiple initialization keys **116** may be generated and used by cryptographic engine **110**, the instant disclosure, for simplicity and conciseness, often refers to a single initialization key **116**. Similarly, although multiple VM keys **114** may be generated per single VM, the instant disclosure, for simplicity and conciseness, may refer to a single VM key **114** per VM **102-j**.

[0019] In some implementations, cryptographic accelerator **110** may communicate with CPU **140** and memory controller **108** via any suitable bus **105**, which may use Compute Express Link™ (CXL) interconnect protocol and/or other interconnect protocols for secure storage of data and data transfers. In some implementations, cryptographic accelerator **110** may be realized in a hardware device having one or more circuits for performing processing operations and a secure memory cache for storing cryptographic keys, intermediate results, final outputs, e.g., as disclosed below in conjunction with FIGS. 2-3.

[0020] Memory controller **108** may further include an EC engine **120** for performing error correction processing. For write (store) memory operations, EC engine **120** may include an EC encoding stage **122** (with implicit key selector encoding) that generates EC bits for both the data being stored and the key selector of the cryptographic key that was used by encryption engine **112** to encrypt the data. For read memory operations, EC engine **120** may include an EC decoding stage **124** (with key selector selection) that generates multiple expanded codewords with various hypothesized key selectors, performs EC decoding for each expanded codeword, and selects the correct key selector hypothesis based on the results of the EC decoding.

[0021] Memory **130** may store VM-shared pages **132** that are encrypted with initialization key **116** and that are accessible to multiple VMs **102-j**. Memory controller **108** may perform read operations of data stored in VM-shared pages **132** after determining that the implicitly stored key selector is that of initialization key **116** and decrypting the data using initialization key **116**. Memory **130** may further store a pool of initialized pages **134** that are encrypted with initialization key **116**. Once an attempt from a specific VM is made to write data to one of the VM-shared pages **132**, one of initialized pages **134** is selected from the pool and allocated to that VM as a VM-allocated page **136**. The new data (encrypted with VM key **114**) may coexist in the VM-allocated page **136** with previously stored data (encrypted with initialization key **116**). When the VM reads a unit of data stored in one of the VM-allocated pages **136**, cryptographic accelerator **110** determines an implicitly stored key selector and decrypts the data. For example, if the implicitly stored key selector is that of initialization key **116** (for the data that has not been modified by the VM), encryption engine **112** applies initialization key **116** to decrypt a unit of data. Conversely, if the implicitly stored key selector is that of VM key **114** (for the data modified by the VM), encryption engine **112** applies VM key **114** to decrypt the unit of data. In both instances, cryptographic accelerator **110** provides the decrypted unit of data to the VM.

[0022] System memory **130** may be (or include) a random-access memory (RAM), electrically erasable programmable read-only memory (EEPROM), flip-flop memory, dynamic random-access memory (DRAM), synchronous DRAM (SDRAM), a static memory, such as static random-access memory (SRAM), and/or the like.

[0023] Example system **100** may further include one or more processors, e.g., CPUs **140**, GPUs, field-programmable gate arrays (FPGA), application-specific integration circuits (ASICs), and so on. “Processor” refers to a device capable of executing instructions encoding arithmetic, logical, or I/O operations. In one illustrative example, a processor may follow Von Neumann architectural model and may include one or more arithmetic logic units (ALUs), a control unit, and may include a cache. In some implementations, some or all of the functions of cryptographic accelerator **110** may be performed by processor **140**. In some implementations, processor **140** and system memory **130** may be implemented as a single controller, e.g., as an FPGA. Processor **140** may be implemented as a single integrated circuit, two or more integrated circuits, or may be a component

of a multi-chip module. Processor **140** may include one or more processor cores. In some implementations, each processor core may execute instructions to run multiple hardware threads, also known as logical processors. One or more logical processors may be assigned to various VMs **102-j**. In some implementations, more than one logical processor may be assigned to a given VM. In some implementations, a single logical processor may be assigned to a group of multiple VMs. [0024] Example system **100** may further include a network interface (not shown in FIG. **1**) to facilitate connection of users' computers that access various VMs **102-j** to example system **100** via one of a variety of networks, e.g., Internet, wireless local area networks (WLAN), personal area networks (PAN), public networks, private networks, and/or the like or any combination thereof. [0025] FIG. **2A** is a schematic diagram illustrating example operations **200** of EC encoding with implicit storage of key selectors using EC data, in accordance with some implementations of the present disclosure. Example operations **200** allow storing data encrypted with different keys in a single memory page, without the need to explicitly store identification of the cryptographic keys that encrypt various portions (units) of the memory page. As illustrated, a VM may attempt to write data **202** to a memory page **210**. Data **202** may have any suitable size, e.g., 512 bytes, 64 bytes, 32 bytes, and/or the like. Data **202** may be encrypted with a VM key for the VM (encryption operations are not explicitly shown in FIG. **2A**). Key selector **204** (e.g., 0 or 1) of the VM key encrypting data **202** may be appended to data **202** to obtain an extended data **206**. Extended data **206** may be processed by EC encoding stage **122** that applies any suitable EC code, e.g., Reed-Solomon code, Bose-Chaudhuri-Hocquenghem code, Goppa code, Hamming code, SECDED code, and/or the like. EC encoding stage **122** may generate EC data **208** for extended data **206**. Data **202** and EC data **208** may then be combined (e.g., concatenated) and stored as codeword **212** in memory page **210**. Key selector **204** is not stored explicitly. Instead, key selector **204** is stored implicitly, encoded in EC data **208** computed in view of key selector **204**.

[0026] FIG. **2B** illustrates schematically memory page **210** that stores data encrypted with different keys, in accordance with some implementations of the present disclosure. As illustrated, memory page **210** may include units of data **214** (depicted with shaded squares) encrypted with a VM key and units of data **216** still encrypted with initialization key (and depicted with white squares). Each unit of data is accompanied with EC data (depicted with cross-hatched squares). Although FIG. **2B** illustrates memory page **210** with 16 units (lines) of stored data, a memory page may store any other number of units of data (e.g., hundreds, thousands, or more units).

[0027] FIG. **2C** is a schematic diagram illustrating example operations **220** of EC decoding with recovery of an implicitly stored key selectors, in accordance with some implementations of the present disclosure. Example operations **220** may be performed to read data previously stored using example operations **200** of FIG. **2A**. In particular, example operations **220** allow reading units of data that are encrypted with different keys and stored in the same memory page, without explicitly stored identification of the cryptographic keys encrypting various units of data. As illustrated, a read operation may be directed to retrieve data stored as part of codeword **212** in memory page **210**. Codeword **212** may be transformed into multiple extended codewords **220-0**, **220-1**, and so on (if $N > 2$ keys need to be identified). Each extended codeword **222-j** includes one or two (or N) possible key selectors **224-j** in a known position. FIG. **2A** and FIG. **2C** illustrate the key selector position that follows all bits of the data (a grey square positioned after white squares), but any other placement of key selectors is also possible, as long as the placement is consistent between EC encoding operations and EC decoding operations. For example, key selector **224-0** having value 0 may correspond to an initialization key **116** and key selector **224-1** having value 1 may correspond to VM key **114** (or vice versa).

[0028] Each extended codeword **222-j** may be processed by a respected stage of the EC decoder, e.g., extended codeword **222-0** may be processed by EC decoding stage **124-0** and extended codeword **222-1** may be processed by EC decoding stage **124-1**. Different EC decoding stages **124-j** may be performed sequentially by the same processing circuits or in parallel by separate

processing circuits. At decision-making block **225**, example operations **220** may include selecting a successful branch of EC decoding. For example, since an incorrect key selector in an extended codeword amounts to an additional error, the successful branch of EC decoding may have fewer errors than the other branch (or $N-1$ other branches, if $N>1$ different key selectors are being disambiguated). In some instances, e.g., if the number of errors in codeword **212** is less than the maximum number t of correctable errors of the specific EC code being used, both data and key selector recovered by each of EC decoding stages **124-j** may be consistent with each other (e.g., may be the same). In particular, the correct key selector corresponds to an EC decoder stage that detects and corrects fewer errors. For example, if the correct key is the initialization key, EC decoding stage **124-1** may detect and correct an error in the position of key selector **224-1** while EC decoding stage **124-0** may detect absence of an error in the position of key selector **224-0**. In those instances where the number of errors in codeword **212** is equal to the maximum number t of correctable errors, an EC decoding stage processing the extended codeword with the correct key selector may succeed to correct errors, while other EC decoding stages (processing extended codewords with $t+1$ errors) may detect an unrecoverable error. (In many instances, even an EC decoding stage processing $t+1$ errors may still correct all errors, using various techniques of error correction beyond the Singleton bound, but this may not be guaranteed in all cases.)

[0029] In some implementations, identifying the correct key selector may be based on computing syndrome values for various extended codewords **222-j**. For example, in Reed-Solomon codes, codewords may be treated as polynomials $CW(x)$ that may be evaluated at a set of roots $x=a.sub.1, a.sub.2 \dots a.sub.k$ (selected as part of EC encoding) and resulting in k syndrome values $S.sub.j=CW(a.sub.j)$. Zero syndrome values, $\{S.sub.j\}=0$, indicate that the retrieved codeword has no error. Some of the syndrome values $S.sub.j$ being non-zero indicate that one or more symbols of the codeword have been corrupted (or that the implicitly stored key selector has been hypothesized incorrectly). Syndrome values $S.sub.j$ may be used as coefficients in a key equation for an error locator polynomial whose solutions identify locations of errors and facilitate identification and correction of the values of those errors.

[0030] Based on the identified key selector, example operations **220** may continue with decrypting the data (which has been corrected of errors, if present) using the identified key selector. For example, if identified $KeyID=0$, decryption may proceed with the initialization key (block **230-0**) whereas identified $KeyID=1$ may cause the decryption to be performed using the VM key (block **230-1**). The decrypted data is then returned to the requesting VM (block **240**) or an application running on the requesting VM, as applicable.

[0031] FIG. 3 illustrates an example architecture of a cryptographic accelerator **300** capable of supporting write and read memory operations using implicitly stored identification of cryptographic keys, in accordance with some implementations of the present disclosure. Cryptographic accelerator **300** combines a write stage **302** that performs storage of data in memory **130** with a read stage **304** that performs retrieval of storage data from memory **130**. Write stage **302** may be capable of performing a write operation requested by a VM (or any other host) as well as initialization of new memory pages (e.g., as may be requested by hypervisor **104** of FIG. 1) to be placed in the pool of initialized memory pages. A Direct Memory Access (DMA) engine **303** may generate memory write operations that involve page initialization. DMA engine **303** may control the type of the memory operation, e.g., by outputting a key selector **306** associated with the operation (or by outputting some other control signal associated with key selector **306**). For example, key selector **306** identifying the initialization key may indicate that a page initialization operation is to be performed while key selector **306** identifying a VM key may indicate that a write operation requested by that VM is to be performed.

[0032] Key selector **306** may serve as a control signal for multiplexers **308**, **314**, and **322**.

Multiplexer **308** may select a correct key, e.g., from VM key **114** and initialization key **116**.

Multiplexer **314** may select a target data, e.g., from write data **310** and initialization data **312**.

(Initialization key **116** and initialization data **312** may be constant values.) In some implementations, initialization data **312** may be zero data (having all zero-bit values). In some implementations, initialization data **312** may be any other predefined data. Multiplexer **322** selects between an address **320** of the write operation or an initialization address **305** for the page being initialized. Encryption stage **316** may encrypt data (e.g., write data **310** or initialization data **312**, depending on the control signal key selector **306**) using the correct key (e.g., VM key **114** or initialization key **116**, depending on key selector **306** provided as the control signal). Encryption stage **316** (which may be a part of encryption engine **112** of FIG. 1) may include an AES block cipher (with a block size of 128 bits, 192 bits, 256 bits), e.g., a 256-bit AES-XTS cipher. In some implementations, encryption stage **316** may include a DES block cipher, IDEA block cipher, Serpent block cipher, Twofish block cipher, and/or other block ciphers. In some implementations, encryption stage **316** may include stream cipher, e.g., Salsa20 stream cipher, Rivest Cipher (RC4), ChaCha stream cipher, and/or other stream ciphers. Encryption stage **316** is followed with EC encoding stage **122** that implicitly stores key selector **306** in EC bits, as disclosed above, e.g., in conjunction with FIG. 2A. A memory controller **325** then stores the encrypted data and EC bits in a memory address of memory **130**, as output by multiplexer **322**.

[0033] Operations of read stage **304** may include receiving an address **320** of a read request and using memory controller **325** to fetch data (codeword) stored at address **320** of memory **130**. The fetched codeword is provided to multiple EC decoding stages, e.g., EC decoding stage **124-0** and EC decoding stage **124-1**. Each EC decoding stage may extend the fetched codeword with a different prospectively assumed value key selector **306** and perform EC decoding of the respective extended codeword, e.g., as disclosed above in conjunction with FIG. 2C. In some implementations, the decoded data outputted by EC decoding stages **124-0** and **124-1** (e.g., Data 0 and Data 1, respectively) may be provided to multiplexer **323**. Syndrome values computed by EC decoding stages **124-0** and **124-1** (e.g., Syn 0 and Syn 1, respectively) may be provided to EC decision stage **327** that determines, based on syndrome values, the correct value of key selector **306**. The determined key selector **306** is used as the control signal provided to multiplexer **323** that selects from Data 0 or Data 1. The control signal with key selector **306** may also be provided to multiplexer **324** that selects the correct key for the selected data, e.g., from initialization key **116** and VM key **114**, which may be stored in key tracker **326**. In some implementations, key tracker **326** may store VM key **114**. For example, when a sequence of read operations is being performed by memory controller **325**, memory controller **325** may execute the operations out-of-order. Correspondingly, key tracker **326** may hold various VM keys **114** associated with read operations pending in memory controller **325** until the respective read operations are completed. Using a correct key provided by multiplexer **324**, a decryption stage **328** decrypts data provided by multiplexer **323**, decryption stage **328** decrypts and outputs read data **330**.

[0034] Multiple variations of the disclosed systems and techniques may also be implemented. The number of EC decoding stages **124-j** may be anywhere between one (where all N possible values of key selector **306** are processed in series) and N (where all possible values of key selector **306** are processed in parallel). For example, if N=4 keys are to be disambiguated, cryptographic accelerator **300** may use two epochs, each epoch processing a pair of possible values of key selector **306** in parallel. Similarly, a cryptographic accelerator with M EC decoding stages may use N/M time epochs, each epoch processing M possible values of key selector **306** in parallel. In some implementations, e.g., in Reed-Solomon EC coded that use 8-bit symbols representing elements of the Galois field $GF(2^{sup.8})$, a single symbol may implicitly encode, using techniques disclosed herein, up to $2^{sup.8}$ different key selectors associated with $2^{sup.8}$ different keys, or, more generally, $2^{sup.p}$ different keys per one element of $GF(2^{sup.p})$. Among $2^{sup.p}$ different keys may be multiple initializations keys used to encrypt separate pools of initialized pages.

[0035] In some instances, less than a full unit of data in a zero-page may have to be stored or updated. In such instances, the partial store/update operation may be performed as a combination of

a read operation and a write operation. More specifically, a read operation may be performed on the full unit of data using the initialization key. A target portion of the unit of data may then be modified/updated and a write operation may be performed using a VM key of the VM performing the update encrypting the full unit of data.

[0036] FIGS. **4-5** are flow diagrams depicting illustrative methods **400-500** of storing and retrieving information with implicitly stored, using EC data, identification of cryptographic keys, in accordance with one or more aspects of the present disclosure. Methods **400-500** and/or each of their individual functions, routines, subroutines, or operations may be performed by a cryptographic accelerator (cryptographic processor), such as cryptographic accelerator **110**, or a general-purpose processor, such as processor **140** of FIG. **1**. Various blocks of methods **400-500** may be performed in a different order compared with the order shown in FIGS. **4-5**. Some operations may be performed concurrently with other operations. Some operations may be optional. Methods **400-500** may be implemented as part of provisioning of trusted VM services protected with memory encryption. In some implementations, at least some operations of methods **400-500** may be performed in parallel, each parallel thread executing one or more individual functions, routines, subroutines, or operations of the methods. In some implementations, parallel threads implementing methods **400-500** may be synchronized (e.g., using semaphores, critical sections, and/or other thread synchronization mechanisms). Alternatively, threads implementing each of methods **400-500** may be executed asynchronously with respect to each other. Various operations of methods **400-500** may be performed in a different order compared with the order shown in FIGS. **4-5**. Some operations of methods **400-500** may be performed concurrently with other operations. In some implementations of methods **400-500**, one or more operations shown in FIGS. **4-5** are not always performed.

[0037] FIG. **4** is a flow diagram of an example method **400** of performing a write operation with implicit storage of cryptographic key identification using error correction data, in accordance with some implementations of the present disclosure. In some implementations, the write operation of method **400** may be performed by one or more processing circuits of write stage **302** of cryptographic accelerator **300** or some other suitable processing device (including but not limited to processor **140** of FIG. **1**). At block **410**, the one or more processing circuits may identify a first data (e.g., data **202** in FIG. **2A**) having a first size. The first data may be encrypted using a cryptographic key (e.g., using encryption stage **316** of FIG. **3**). In some implementations, the cryptographic key may be an ephemeral key. As indicated with block **412**, the cryptographic key may be selected (e.g., using multiplexer **308** of FIG. **3**) from a group of cryptographic keys that may include a first cryptographic key encrypting data shared by a plurality of hosts (e.g., an initialization key **116**) and a second (third, etc.) cryptographic key encrypting private data associated with a first (second, etc.) host of the plurality of hosts. In some implementations, the first (second, etc.) host of the plurality of hosts may be or include a virtual machine, and the first cryptographic key may, correspondingly, be (or include) a VM key **114** for the respective virtual machine.

[0038] At block **420**, method **400** may continue with the one or more processing circuits generating a second data (e.g., extended data **206** in FIG. **2A**) that includes the first data and a key selector of the cryptographic key.

[0039] At block **430**, the one or more processing circuits performing method **400** may generate an error correction (EC) data (e.g., EC data **208**) using the second data (e.g., using EC encoding stage **122** of FIG. **3**). The EC data may have a second size (which may or may not be different from the first size). In some implementations, to generate the EC data, the one or more processing circuits are to apply a Reed-Solomon EC code to the second data.

[0040] At block **440**, method **400** may include storing a third data (e.g., codeword **212** in FIG. **2A**) in a memory (e.g., memory **130**). The third data may include the first data and the EC data but exclude the key selector. In some implementations, the third size does not exceed a sum of the first

size and the second size. The third data may be stored in a memory partition (e.g., memory page of FIGS. 2A-2C) storing a plurality of memory units, wherein a first subset of one or more memory units of the plurality of memory units is encrypted with the cryptographic key (e.g., VM key **114** or initialization key **116**) and a second subset of one or more memory units of the plurality of memory units is encrypted with a different cryptographic key (e.g., initialization key **116** or VM key **114**). [0041] FIG. 5 is a flow diagram of an example method **500** of performing a read operation with recovery of implicitly stored, using error correction data, cryptographic key identification, in accordance with some implementations of the present disclosure. In some implementations, the read operation of method **500** may be performed by one or more processing circuits of read stage **304** of cryptographic accelerator **300** or some other suitable processing device (including but not limited to processor **140** of FIG. 1). At block **510**, the one or more processing circuits may retrieve a codeword (e.g., codeword **212** in FIG. 2C) from a memory (e.g., memory **130**). The codeword may include a first data encrypted with a cryptographic key and an EC data.

[0042] At block **520**, method **500** may continue with the one or more processing circuits identifying, using the EC data, a key selector of the cryptographic key. In some implementations, to identify the key selector, the one or more processing circuits may apply a Reed-Solomon EC code to the codeword. As illustrated with the callout portion of FIG. 5, identifying the key selector may include, at block **522**, forming a plurality of extended codewords (e.g., extended codewords **222-0**, **220-1**, etc.). Each of the plurality of extended codewords may be formed by combining the codeword with one of a plurality of key selectors (e.g., key selectors **224-0**, **224-1**, etc.) identifying a respective one of the plurality of cryptographic keys. At block **524**, method **500** may include applying an EC code to each of the plurality of extended codewords (e.g., via ED decoding stages **124-0**, **124-1**, etc., in FIG. 2A and FIG. 3). In some implementations, applying the EC code to each of the plurality of extended codewords may include computing syndrome values for each of the plurality of extended codewords and identifying the key selector of the cryptographic key using the computed syndrome values. In some implementations, applying the EC code to each of the plurality of extended codewords may include identifying a target extended codeword of the plurality of extended codewords, the target extended codeword having fewer errors than each other of the plurality of extended codewords. The correct key selector may then be identified as a key selector included in the target extended codewords. In some implementations, the one or more processing circuits are to apply the EC code to at least two of the plurality of extended codewords in parallel (e.g., as illustrated with EC decoding stages **124-1** and **124-2** in FIG. 3).

[0043] At block **530**, the one or more processing circuits performing method **500** may select, using the key selector, the cryptographic key from the plurality of cryptographic keys (e.g., using multiplexers **323**, **324**, and EC decision stage **327** in FIG. 3) and, at block **540**, may decrypt the first data using the cryptographic key (e.g., using decryption stage **328** in FIG. 3).

[0044] FIG. 6 depicts a block diagram of an example computer system **600** operating in accordance with one or more aspects of the present disclosure. In various illustrative examples, example computer system **600** may include example system **100**, illustrated in FIG. 1. Example computer system **600** may be connected to other computer systems in a LAN, an intranet, an extranet, and/or the Internet. Computer system **600** may operate in the capacity of a server in a client-server network environment. Computer system **600** may be a personal computer (PC), a set-top box (STB), a server, a network router, switch or bridge, or any device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that device. Further, while only a single example computer system is illustrated, the term “computer” shall also be taken to include any collection of computers that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methods discussed herein.

[0045] Example computer system **600** may include a processing device **602** (also referred to as a processor or CPU), a main memory **604** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM), etc.), a static memory

606 (e.g., flash memory, static random access memory (SRAM), etc.), and a secondary memory (e.g., a data storage device **618**), which may communicate with each other via a bus **630**.

[0046] Processing device **602** represents one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. More particularly, processing device **602** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **602** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. In accordance with one or more aspects of the present disclosure, processing device **602** may be configured to execute instructions implementing method **400** of performing a write operation with implicit storage of cryptographic key identification using error correction data, and method **500** of performing a read operation with recovery of implicitly stored, using error correction data, cryptographic key identification.

[0047] Example computer system **600** may further comprise a network interface device **608**, which may be communicatively coupled to a network **620**. Example computer system **600** may further comprise a video display **610** (e.g., a liquid crystal display (LCD), a touch screen, or a cathode ray tube (CRT)), an alphanumeric input device **612** (e.g., a keyboard), a cursor control device **614** (e.g., a mouse), and an acoustic signal generation device **616** (e.g., a speaker).

[0048] Data storage device **618** may include a computer-readable storage medium (or, more specifically, a non-transitory computer-readable storage medium) **626** on which is stored one or more sets of executable instructions **622**. In accordance with one or more aspects of the present disclosure, executable instructions **622** may comprise executable instructions implementing method **400** of performing a write operation with implicit storage of cryptographic key identification using error correction data, and method **500** of performing a read operation with recovery of implicitly stored, using error correction data, cryptographic key identification.

[0049] Executable instructions **622** may also reside, completely or at least partially, within main memory **604** and/or within processing device **602** during execution thereof by example computer system **600**, main memory **604** and processing device **602** also constituting computer-readable storage media. Executable instructions **622** may further be transmitted or received over a network via network interface device **608**.

[0050] While the computer-readable storage medium **626** is shown in FIG. 6 as a single medium, the term “computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of operating instructions. The term “computer-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine that cause the machine to perform any one or more of the methods described herein. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

[0051] Some portions of the detailed descriptions above are presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0052] It should be borne in mind, however, that all of these and similar terms are to be associated

with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise, as apparent from the following discussion, it is appreciated that throughout the description, discussions utilizing terms such as “identifying,” “determining,” “storing,” “adjusting,” “causing,” “returning,” “comparing,” “creating,” “stopping,” “loading,” “copying,” “throwing,” “replacing,” “performing,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0053] Examples of the present disclosure also relate to an apparatus for performing the methods described herein. This apparatus may be specially constructed for the required purposes, or it may be a general-purpose computer system selectively programmed by a computer program stored in the computer system. Such a computer program may be stored in a computer readable storage medium, such as, but not limited to, any type of disk including optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic disk storage media, optical storage media, flash memory devices, other type of machine-accessible storage media, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0054] The methods and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear as set forth in the description below. In addition, the scope of the present disclosure is not limited to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the present disclosure.

[0055] It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other implementation examples will be apparent to those of skill in the art upon reading and understanding the above description. Although the present disclosure describes specific examples, it will be recognized that the systems and methods of the present disclosure are not limited to the examples described herein, but may be practiced with modifications within the scope of the appended claims. Accordingly, the specification and drawings are to be regarded in an illustrative sense rather than a restrictive sense. The scope of the present disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A processor to: identify a first data having a first size, wherein the first data is encrypted using a cryptographic key; generate a second data comprising the first data and a key selector of the cryptographic key; generate an error correction (EC) data using the second data, the EC data having a second size; and store a third data in a memory, the third data having a third size, wherein the third data comprises the first data and the EC data, and wherein the third size does not exceed a sum of the first size and the second size.
2. The processor of claim 1, wherein the cryptographic key is selected from a group of cryptographic keys comprising: a first cryptographic key encrypting data shared by a plurality of hosts; and a second cryptographic key encrypting private data associated with a first host of the plurality of hosts.
3. The processor of claim 2, wherein the group of keys further comprises: a third cryptographic key

encrypting private data associated with a second host of the plurality of hosts.

4. The processor of claim 2, wherein the first host of the plurality of hosts comprises a virtual machine.

5. The processor of claim 1, wherein to generate the EC data, the processor is to apply a Reed-Solomon EC code to the second data.

6. The processor of claim 1, wherein the third data is stored in a memory partition storing a plurality of memory units, wherein a first subset of one or more memory units of the plurality of memory units is encrypted with the cryptographic key and a second subset of one or more memory units of the plurality of memory units is encrypted with a different cryptographic key.

7. A processor to: retrieve a codeword from a memory, the codeword comprising: a first data encrypted with a cryptographic key, and an EC data; identify, using the EC data, a key selector of the cryptographic key; select, using the key selector, the cryptographic key from a plurality of cryptographic keys; and decrypt the first data using the cryptographic key.

8. The processor of claim 7, wherein to identify the key selector, the processor is to: form a plurality of extended codewords, wherein each of the plurality of extended codewords is formed by combining the codeword with one of a plurality of key selectors identifying a respective one of the plurality of cryptographic keys; and apply an EC code to each of the plurality of extended codewords.

9. The processor of claim 8, wherein to apply the EC code to each of the plurality of extended codewords, the processor is to: compute syndrome values for each of the plurality of extended codewords; and identify the key selector of the cryptographic key using the computed syndrome values.

10. The processor of claim 8, wherein to apply the EC code to each of the plurality of extended codewords, the processor is to: identify a target extended codeword of the plurality of extended codewords, wherein the target extended codeword comprises fewer errors than each other of the plurality of extended codewords; and identify the key selector as a key selector included in the target extended codewords.

11. The processor of claim 8, wherein the processor is to apply the EC code to at least two of the plurality of extended codewords in parallel.

12. The processor of claim 7, wherein the key selector identifies the cryptographic key from a group of cryptographic keys comprising: a first cryptographic key encrypting data shared by a plurality of hosts; and a second cryptographic key encrypting private data associated with a first host of the plurality of hosts.

13. The processor of claim 11, wherein the group of keys further comprises: a third cryptographic key encrypting private data shared of a second host of the plurality of hosts.

14. The processor of claim 11, wherein the first host of the plurality of hosts comprises a virtual machine.

15. The processor of claim 7, wherein to identify the key selector, the processor is to apply a Reed-Solomon EC code to the codeword.

16. The processor of claim 7, wherein the codeword is stored in a memory partition storing a plurality of memory units, wherein a first subset of one or more memory units of the plurality of memory units is encrypted with the cryptographic key and a second subset of one or more memory units of the plurality of memory units is encrypted with a different cryptographic key.

17. A processing device comprising: a first plurality of circuits to: encrypt a first data using a cryptographic key of a plurality of cryptographic keys; generate a second data comprising the first data and a key selector of the cryptographic key; generate an error correction (EC) data using the second data; and store a codeword in a memory, wherein the codeword comprises the first data and the EC data; and a second plurality of circuits to: retrieve the codeword from the memory; identify, using the EC data, the key selector; select, using the key selector, the cryptographic key from the plurality of cryptographic keys; and decrypt the first data using the cryptographic key.

- 18.** The processing device of claim 17, processor of claim 1, wherein the first plurality of circuits is to select the cryptographic key from at least: a first cryptographic key encrypting data shared by a plurality of hosts; and a plurality of cryptographic keys encrypting private data associated with a respective host of the plurality of hosts.
- 19.** The processing device of claim 17, wherein to identify the key selector, the second plurality of circuits is to: form a plurality of extended codewords, wherein each of the plurality of extended codewords is formed by combining the codeword with one of a plurality of key selectors identifying a respective one of the plurality of key selectors; and apply an EC code to each of the plurality of extended codewords.
- 20.** The processing device of claim 19, wherein the second plurality of circuits are to apply the EC code to at least two of the plurality of extended codewords in parallel.
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